

End-to-End Analysis Case Study

Capstone Project — Week 4

Project Objective

This project follows the required end-to-end workflow: clean and explore the dataset in Python, analyze it using Pandas, visualize the results using Excel and Power BI, and communicate the findings through business recommendations.

Executive KPI Summary

KPI	Result
Total Sales	\$118.73M
Total Profit	\$16.89M
Profit Margin	14.23%
Units Sold	1.13M
Discount Rate	7.20%

1. Methodology

The submitted Python notebook imports the dataset with Pandas, inspects the structure and data types, checks missing values and duplicates, cleans column names, handles missing Discount Band values, converts Date to datetime, and creates Profit Margin and Discount Rate fields. It then performs grouped analysis by country, product, segment, year and discount band.

Final Cleaned_Data sheet: 700 rows × 18 columns; no missing values and no duplicate rows.

2. Key Findings

- United States has the highest sales at \$25.03M; France has the highest total profit at \$3.78M.
- Paseo is the leading product by sales and profit, with \$33.01M sales and \$4.80M profit.
- Enterprise produces \$19.61M sales but a loss of \$0.61M.
- 2014 sales grew by approximately 249.5% versus 2013, while profit grew by approximately 235.6%.
- Higher discount bands are associated with lower profitability; this is an association rather than a causal conclusion.

3. Power BI Dashboard

Finally, I built an interactive dashboard in Power BI: Executive Overview, Performance Analysis, and Business Insights.

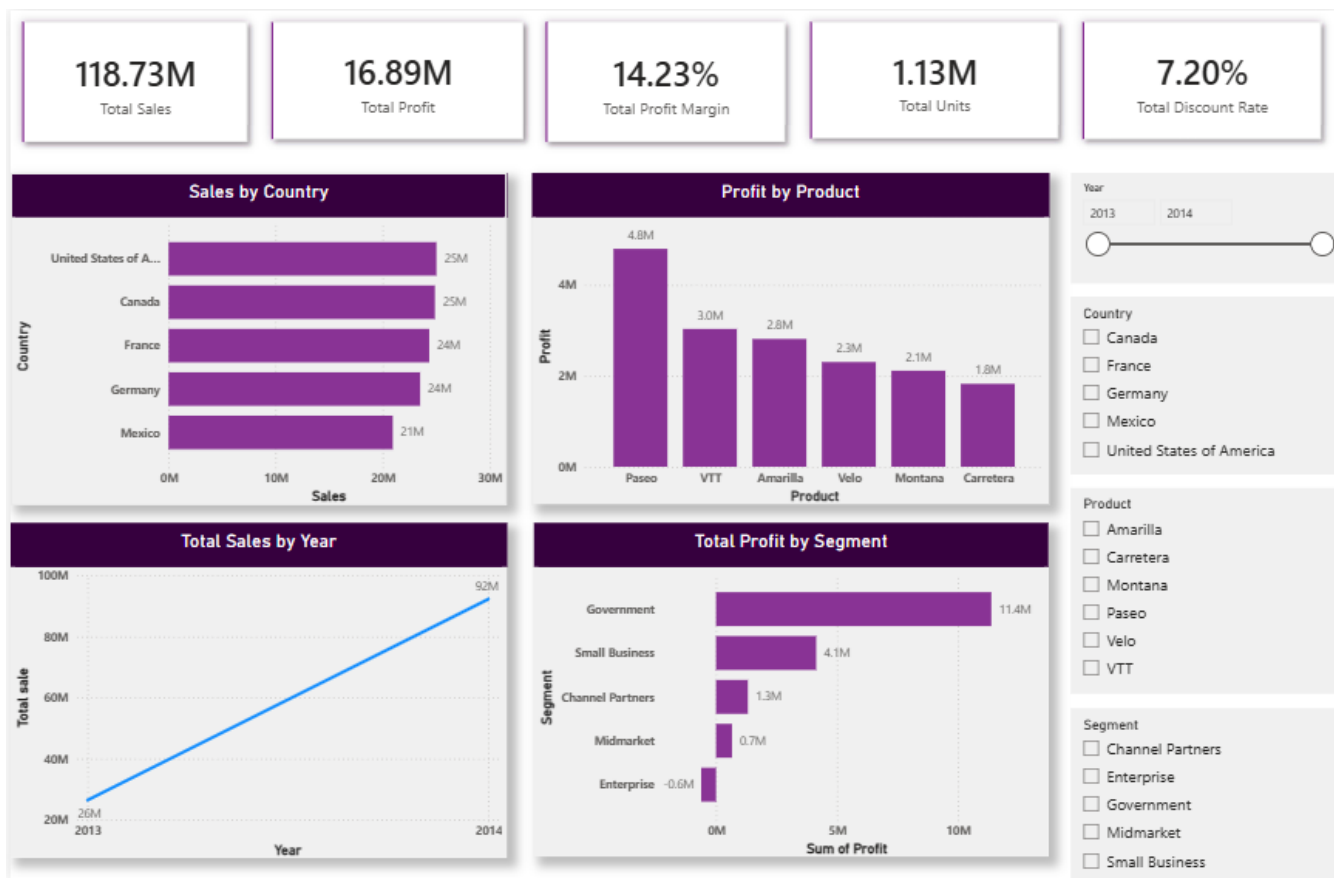


Figure 1. Executive Overview

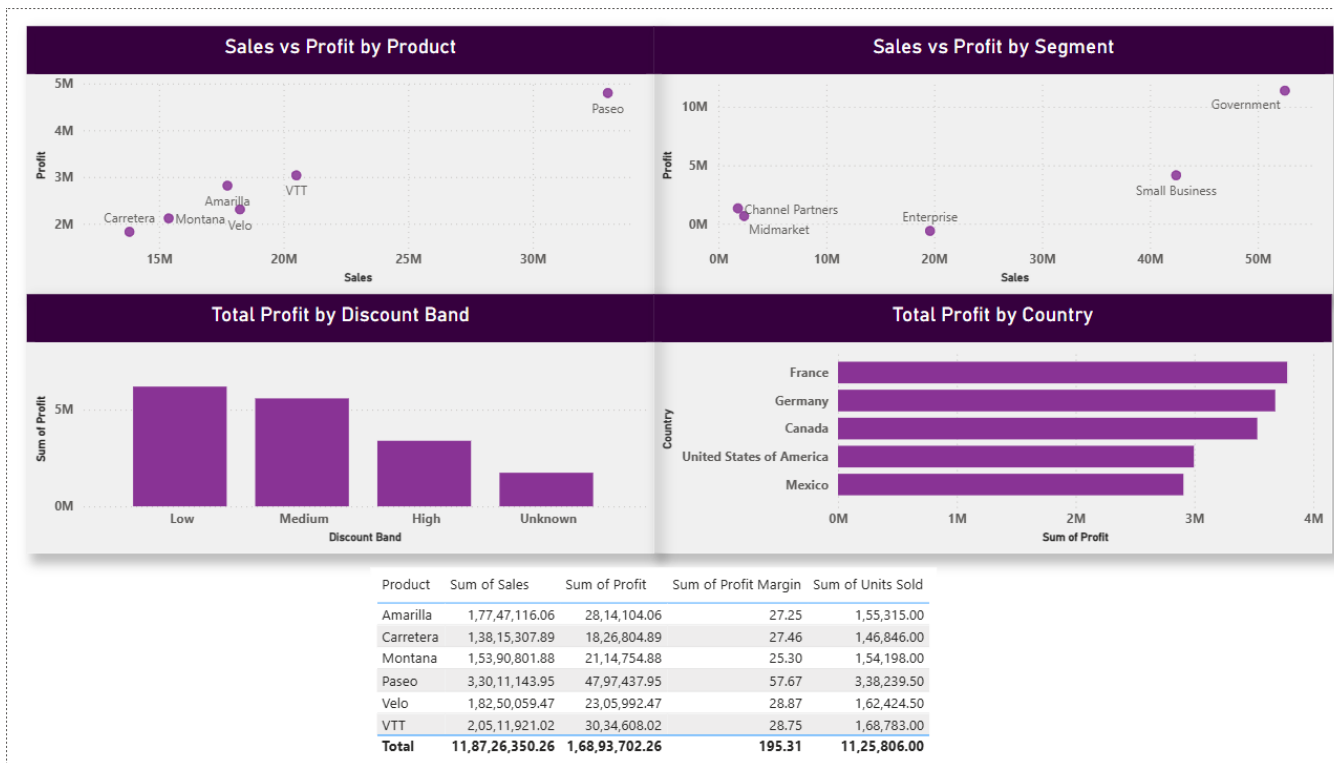


Figure 2. Performance Analysis

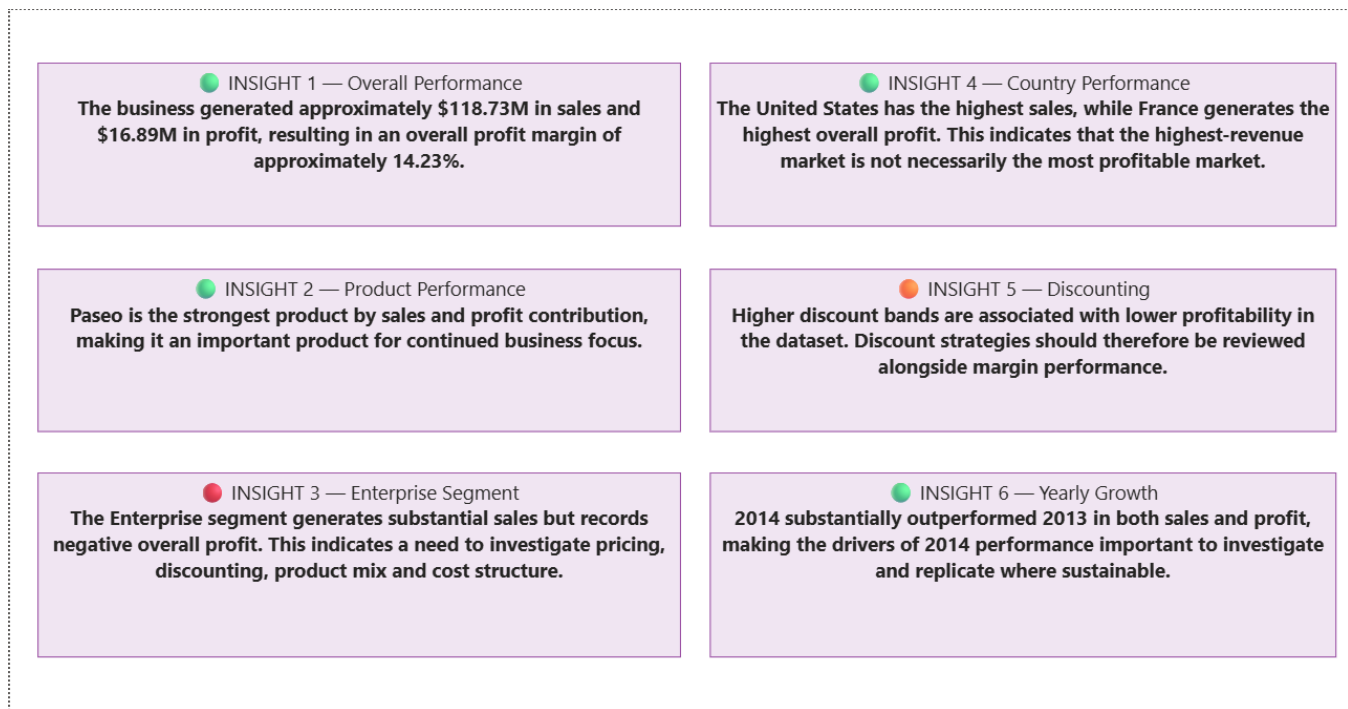


Figure 3. Business Insights & Recommendations

4. Business Recommendations

1. Investigate Enterprise pricing, discounting, product mix and cost structure.
2. Protect and expand high-performing products, particularly Paseo.
3. Optimize discounts using margin performance as a key constraint.
4. Study France's profitability drivers and transfer suitable practices.
5. Investigate the products, countries and segments driving 2014 growth.

5. Conclusion

The project provides an end-to-end analytics workflow from Python cleaning and exploration through Pandas analysis and Power BI reporting. The main management opportunity is to improve profit quality while sustaining the strong growth and product performance shown in the analysis.

Appendix — Key Python Code

```
# Cleaning

df.columns = df.columns.str.strip()

df["Discount Band"] = df["Discount Band"].fillna("Unknown")

df["Date"] = pd.to_datetime(df["Date"], errors="coerce")


# Calculated fields

df["Profit Margin"] = np.where(

    df["Sales"] != 0, df["Profit"] / df["Sales"], np.nan

)

df["Discount Rate"] = np.where(

    df["Gross Sales"] != 0, df["Discounts"] / df["Gross Sales"], np.nan

)


# Grouped analysis

country_analysis = df.groupby("Country").agg(

    Sales=("Sales", "sum"),

    Profit=("Profit", "sum"),

    Units_Sold=("Units Sold", "sum")

).reset_index()


product_analysis = df.groupby("Product").agg(

    Sales=("Sales", "sum"),

    Profit=("Profit", "sum"),

    Units_Sold=("Units Sold", "sum")

).reset_index()


segment_analysis = df.groupby("Segment").agg(

    Sales=("Sales", "sum"),

    Profit=("Profit", "sum"),

    Units_Sold=("Units Sold", "sum")

).reset_index()
```

```
year_analysis = df.groupby("Year").agg(  
    Sales=("Sales", "sum"),  
    Profit=("Profit", "sum"),  
    Units_Sold=("Units Sold", "sum")  
) .reset_index()
```